

United Kingdom 2017

1 Survey Description

Survey: Understanding Society (2017)

Link to the document: <https://www.understandingsociety.ac.uk/about/british-household-panel-survey/>

Sample: Understanding Society is a longitudinal, household panel, multi-scope study conducted by the Institute for Social and Economic Research at the University of Essex. The Study began in 2009 as a continuation of the British Household Panel Study (BHPS), which ran from 1991 to 2008. The Understanding Society sample design consists of four components: the General Population Sample (GPS), the Ethnic Minority Boost Sample (EMBS), the former British Household Panel Survey (BHPS) sample, and the Immigrant and Ethnic Minority Boost Sample (IEMBS). Understanding Society employs a probability sampling approach with a complex sample design, incorporating clustering and stratification, and includes unequal selection probabilities to ensure representation of population subgroups. The GPS uses a proportionately stratified, clustered sample of residential addresses from the Postcode Address File in England, Scotland, and Wales, and an unclustered systematic random sample from the Land and Property Services Agency list in Northern Ireland.. There are 34,864 individuals in the total sample and 10,180 individuals in the analysis sample. Section 3 of this document describes the prevalence and pattern of missing data.

Weights: Understanding Society employs a probability sampling approach with a complex sample design, incorporating clustering and stratification, and includes unequal selection probabilities to ensure representation of population subgroups. The GPS uses a proportionately stratified, clustered sample of residential addresses from the Postcode Address File in England, Scotland, and Wales, and an unclustered systematic random sample from the Land and Property Services Agency list in Northern Ireland. The weights account for this stratification.

Outcome: The outcome variables are annual equivalized household disposable total income in dollars PPP 2017.¹

Circumstances:

- Race/Ethnicity (described in Table 1)
- Sex (female, male)
- Country of Birth (24 categories described in Table 1)
- Fathers's edu. (4 levels of education, described in Table 2)
- Mother's edu. (4 levels of education, described in Table 2)
- Father's occ. (8 categories NSSEC, described in Table 3)
- Mother's occ. (8 categories NSSEC, described in Table 3)

¹Income variable was equivalized using the square root scale.

2 Descriptive Statistics

Table 1: Respondant's socio-demographics - 2017

	Analysis sample	Total sample
	(N=10,180)	(N=34,864)
Gender		
0 Male	4,271 (42.0%)	15,444 (44.3%)
1 Female	5,909 (58.0%)	19,420 (55.7%)
Ethnicity		
1 Black	351 (3.4%)	1,403 (4.0%)
2 White	9,198 (90.4%)	28,392 (81.4%)
3 Asian	435 (4.3%)	3,886 (11.1%)
4 Mixed	154 (1.5%)	638 (1.8%)
6 Other	42 (0.4%)	244 (0.7%)
Missing	0 (0%)	301 (0.9%)

Table 2: Parental education - 2017

	Analysis sample	Total sample
	(N=10,180)	(N=34,864)
Mother's education (levels)		
1 Primary	3,940 (38.7%)	11,334 (32.5%)
2 Secondary Lower	3,144 (30.9%)	6,846 (19.6%)
3 Secondary Upper	2,112 (20.7%)	4,044 (11.6%)
4 Tertiary	984 (9.7%)	1,734 (5.0%)
Missing	0 (0%)	10,906 (31.3%)
Father's education (levels)		
1 Primary	3,832 (37.6%)	9,644 (27.7%)
2 Secondary Lower	2,245 (22.1%)	4,924 (14.1%)
3 Secondary Upper	2,847 (28.0%)	5,861 (16.8%)
4 Tertiary	1,256 (12.3%)	2,632 (7.5%)
Missing	0 (0%)	11,803 (33.9%)

Table 3: Parental occupation - 2017

	Analysis sample	Total sample
	(N=10,180)	(N=34,864)
Mother's occupation		
1 Managers	496 (4.9%)	1,295 (3.7%)
2 High Professional	782 (7.7%)	1,852 (5.3%)
3 Low Professional	1,660 (16.3%)	3,841 (11.0%)
4 Intermediate	1,059 (10.4%)	2,433 (7.0%)
5 Seftemployed	1,466 (14.4%)	4,127 (11.8%)
6 Technical	1,571 (15.4%)	3,671 (10.5%)
7 Semi routine	1,343 (13.2%)	3,313 (9.5%)
8 Routine	1,803 (17.7%)	4,730 (13.6%)
Missing	0 (0%)	9,602 (27.5%)
Father's occupation		
1 Managers	107 (1.1%)	160 (0.5%)
2 High Professional	175 (1.7%)	298 (0.9%)
3 Low Professional	2,067 (20.3%)	3,169 (9.1%)
4 Intermediate	1,961 (19.3%)	3,024 (8.7%)
5 Seftemployed	652 (6.4%)	1,069 (3.1%)
6 Technical	242 (2.4%)	346 (1.0%)
7 Semi routine	2,619 (25.7%)	4,387 (12.6%)
8 Routine	2,357 (23.2%)	3,946 (11.3%)
Missing	0 (0%)	18,465 (53.0%)

Table 4: Respondant's income - 2017

	N	Mean	SD	Median	Min	Max	Missing
Analysis sample	10,180	55,622	121,911	47,117	1.231	12,011,321	0
Total sample	34,864	52,988	151,880	43,440	1.231	23,947,256	865

3 Missing data analysis

3.1 Missing patterns

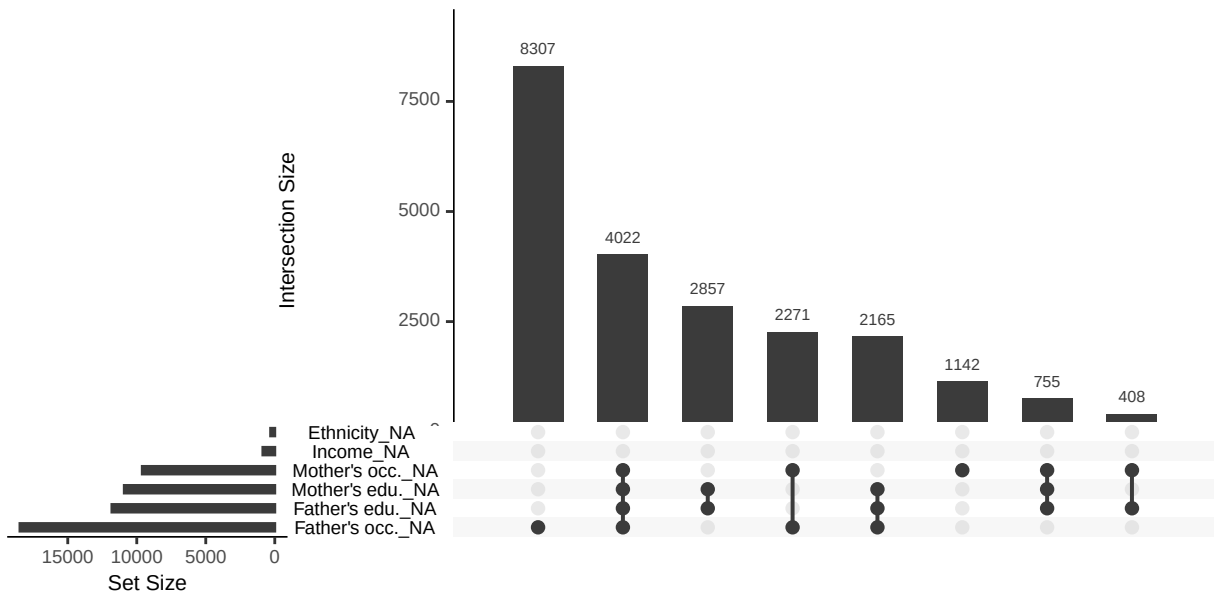


Figure 1: Missing patterns: *Left*: Marginal distribution of missing observations per variable. *Right*: Combination of missingness across cases

3.2 Differences in expected total equivalized household income between samples

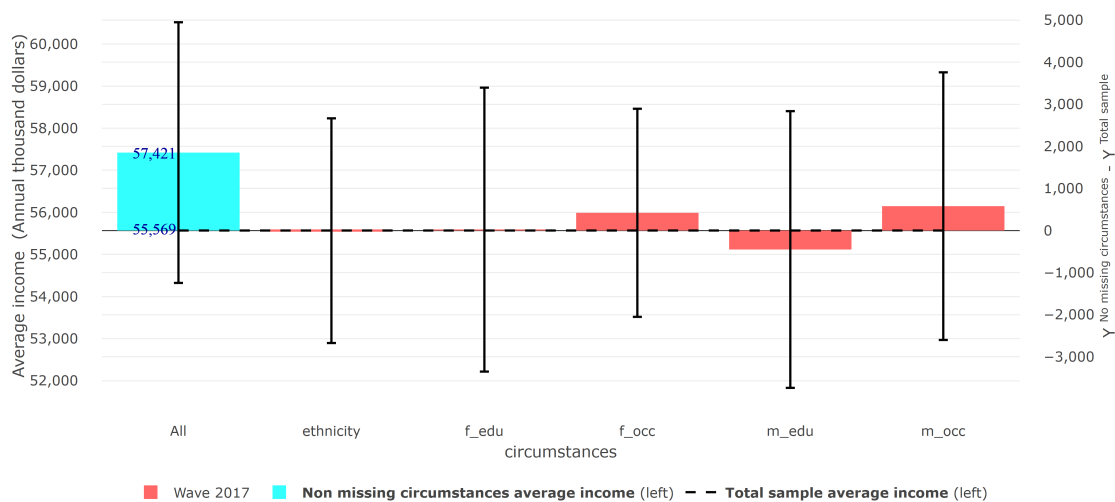


Figure 2: Differences in expected total equivalized household income between the sample with non-missing circumstances and the total sample

3.3 Gini coefficient

Table 5: Gini coefficient in analysis sample and total sample

Wave	Sample	Gini	Lower bound	Upper bound	Average income
Wave 2017	Analysis sample	0.354	0.323	0.38	57,421
Wave 2017	Total sample	0.368	0.339	0.39	55,569

3.4 Differences in Gini coefficient between samples

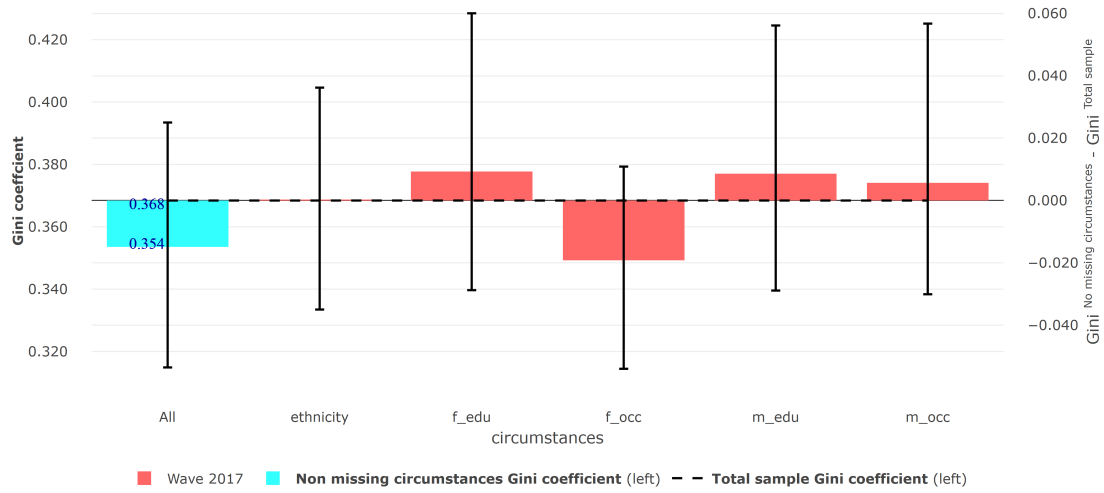


Figure 3: Differences in Gini coefficient between the sample with non-missing circumstances and the total sample